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Deploy a React app with Node.js

In this tutorial you will learn how to build and deploy your React app using an Express.js server.

With this setup you can deploy your React app on Heroku with little effort.

Of course with little configuration you can deploy it to other providers as well.



[You can find the project source code on github here](#)

Here is a brief plan of how you can do it fast and easy:

Tools needed

- node.js
- express.js
- create-react-app
- VS Code or anything you like

Todo List

- Installation of dependencies
- Setup the client
- Build the server

Installation

[Install node if you don't have it already](#)

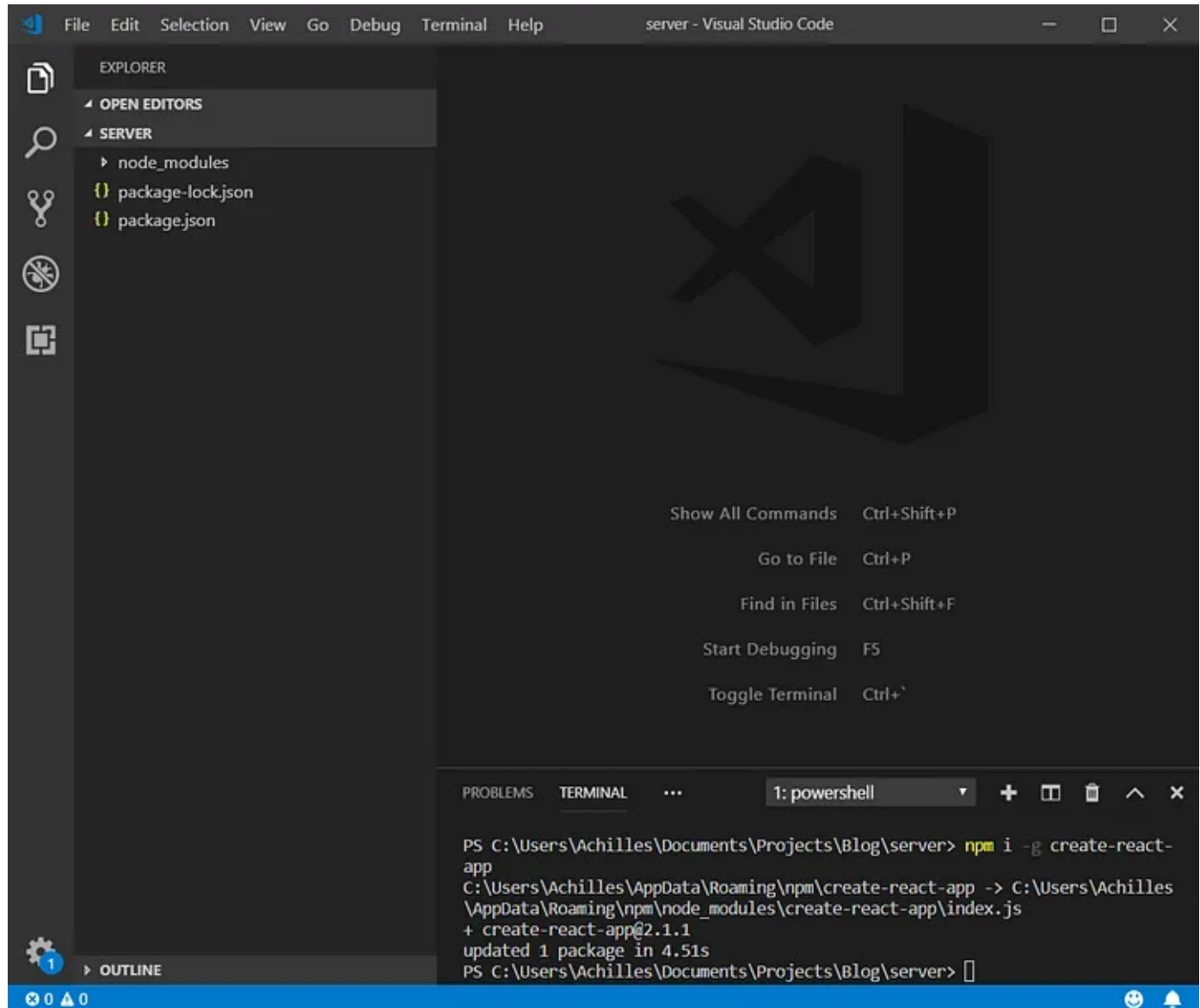
My version is 8.12.0

Assuming you are in the folder you want to create your project run in your terminal:

```
mkdir server
cd server
npm init -y
npm install express --save
npm i -g create-react-app
```

With the above commands , you initialize your project and install the dependencies for the server and the react client.

Your project will look like this



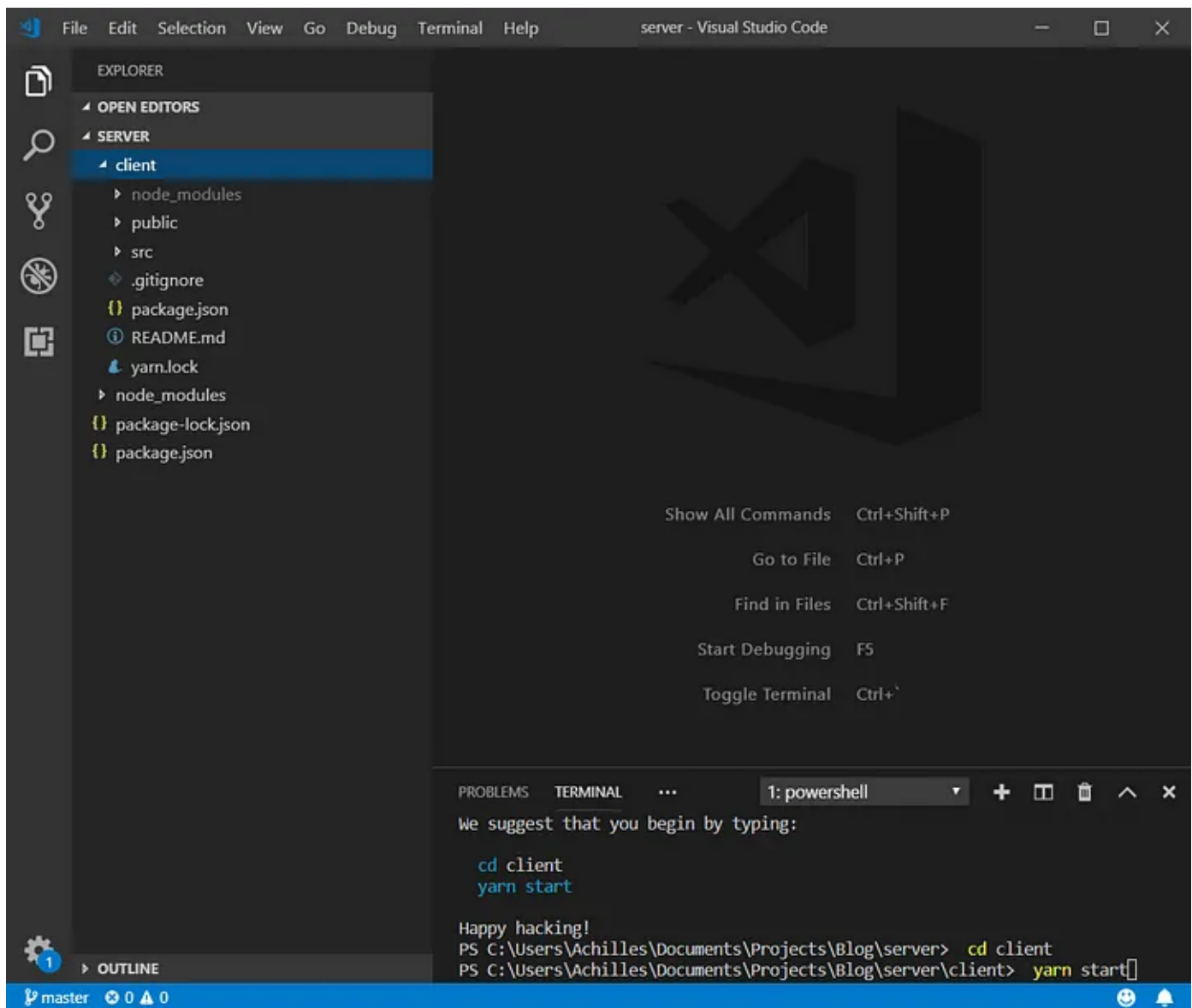
Setup the client

Now you have everything you need to setup your react project, lets start with the client.

on your terminal run :

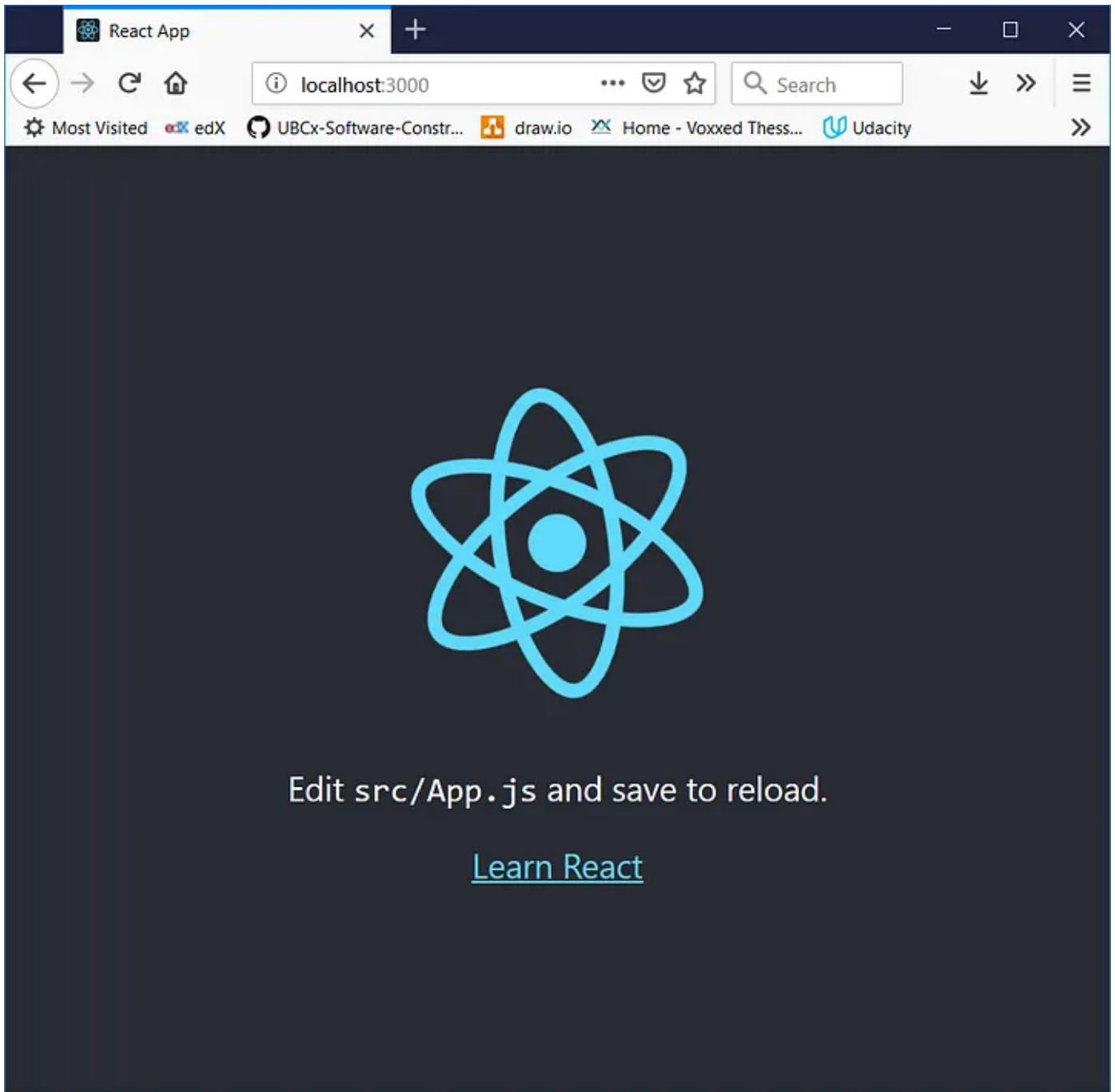
```
create-react-app client
cd client
yarn start
```

Note that 'client' is the name of the react app , you can pick a different name if you like.



Wait for the process to complete.

Now you can see the react app in your browser served by the development server



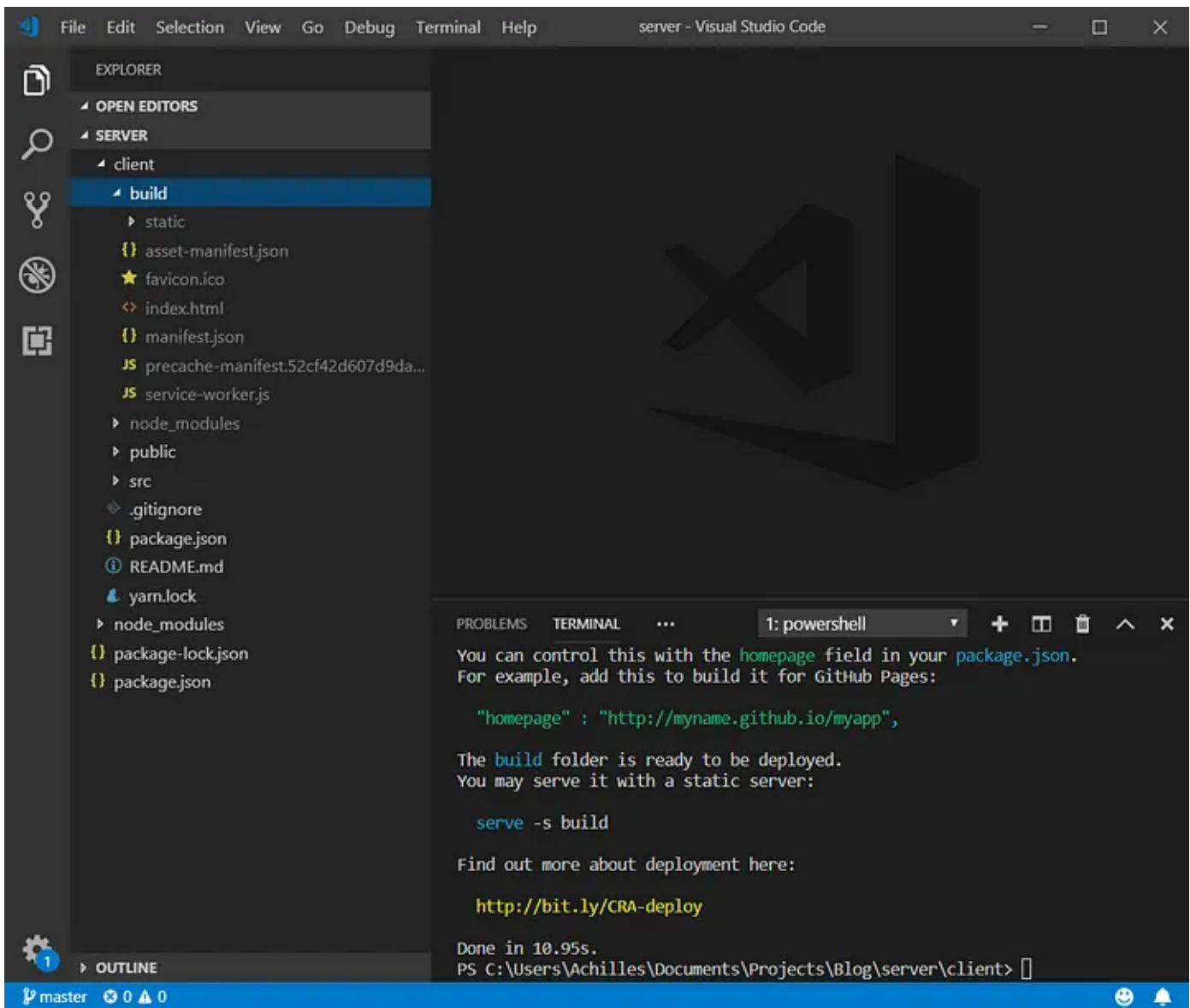
The development server provided by create-react-app is on port 3000

press **Ctrl + C** on the terminal to close the development server

Build your project by running :

```
yarn build
```

Wait for the process to end , the react code for production lives in the **build** directory.



go back to your project root folder :

```
cd ..
```

Building the server

Now you will develop the **production server** of your react app.

*You will need a production server to serve the react app over the internet, the **development server** included with **create-react-app** is only used for development purposes!*

Create a new file , name it **index.js** and put the following:

```
const express = require('express');
const app = express();
// serve up production assets
app.use(express.static('client/build'));
// let the react app to handle any unknown routes
// serve up the index.html if express doesn't recognize the route
const path = require('path');
app.get('*', (req, res) => {
  res.sendFile(path.resolve(__dirname, 'client', 'build',
    'index.html'));
});
```

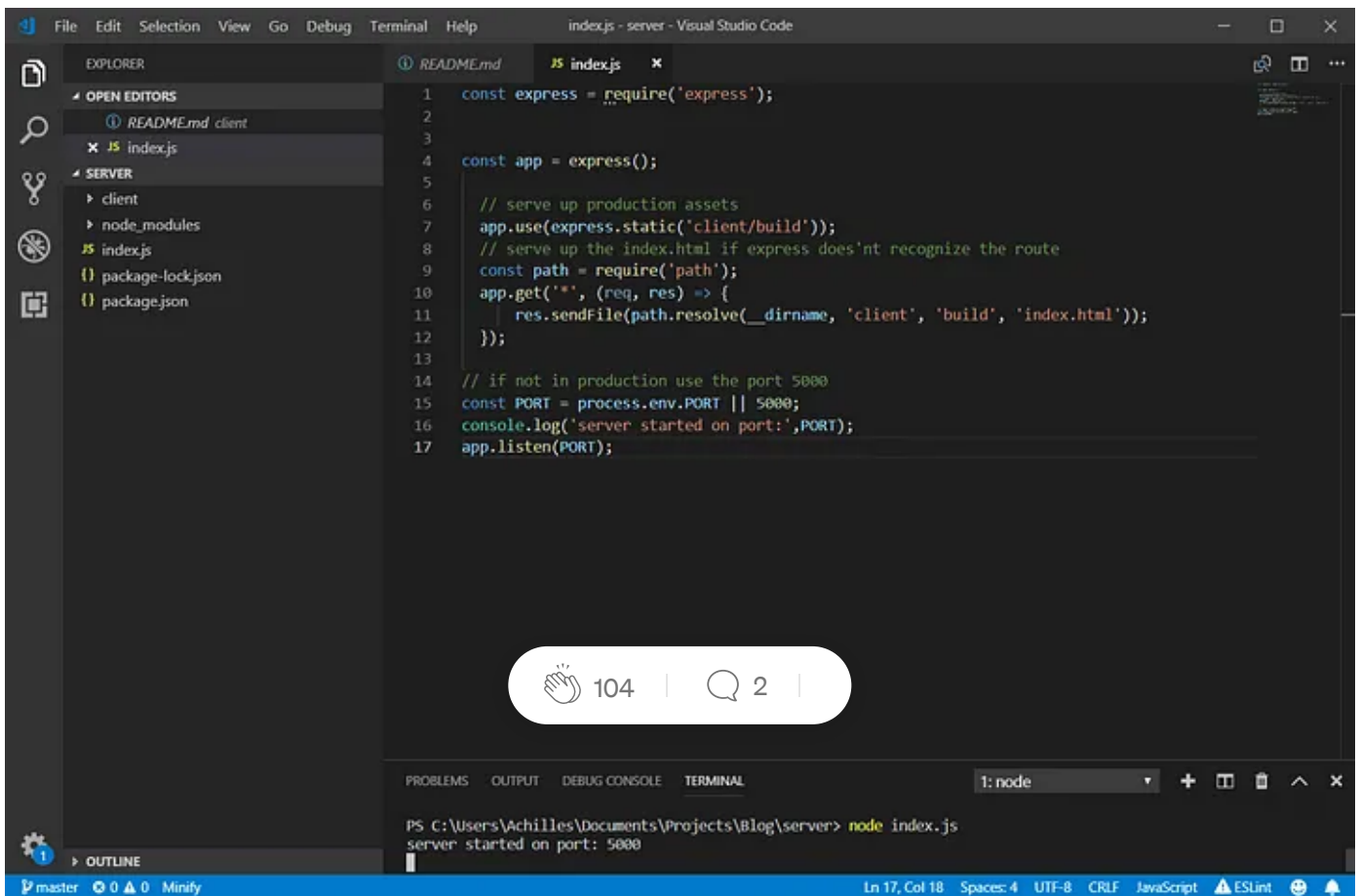
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```
console.log('server started on port:',PORT);
app.listen(PORT);
```

save the file.

run on your terminal:

```
node index.js
```



```
1 const express = require('express');
2
3
4 const app = express();
5
6 // serve up production assets
7 app.use(express.static('client/build'));
8 // serve up the index.html if express doesn't recognize the route
9 const path = require('path');
10 app.get('*', (req, res) => {
11   res.sendFile(path.resolve(__dirname, 'client', 'build', 'index.html'));
12 });
13
14 // if not in production use the port 5000
15 const PORT = process.env.PORT || 5000;
16 console.log('server started on port:', PORT);
17 app.listen(PORT);
```

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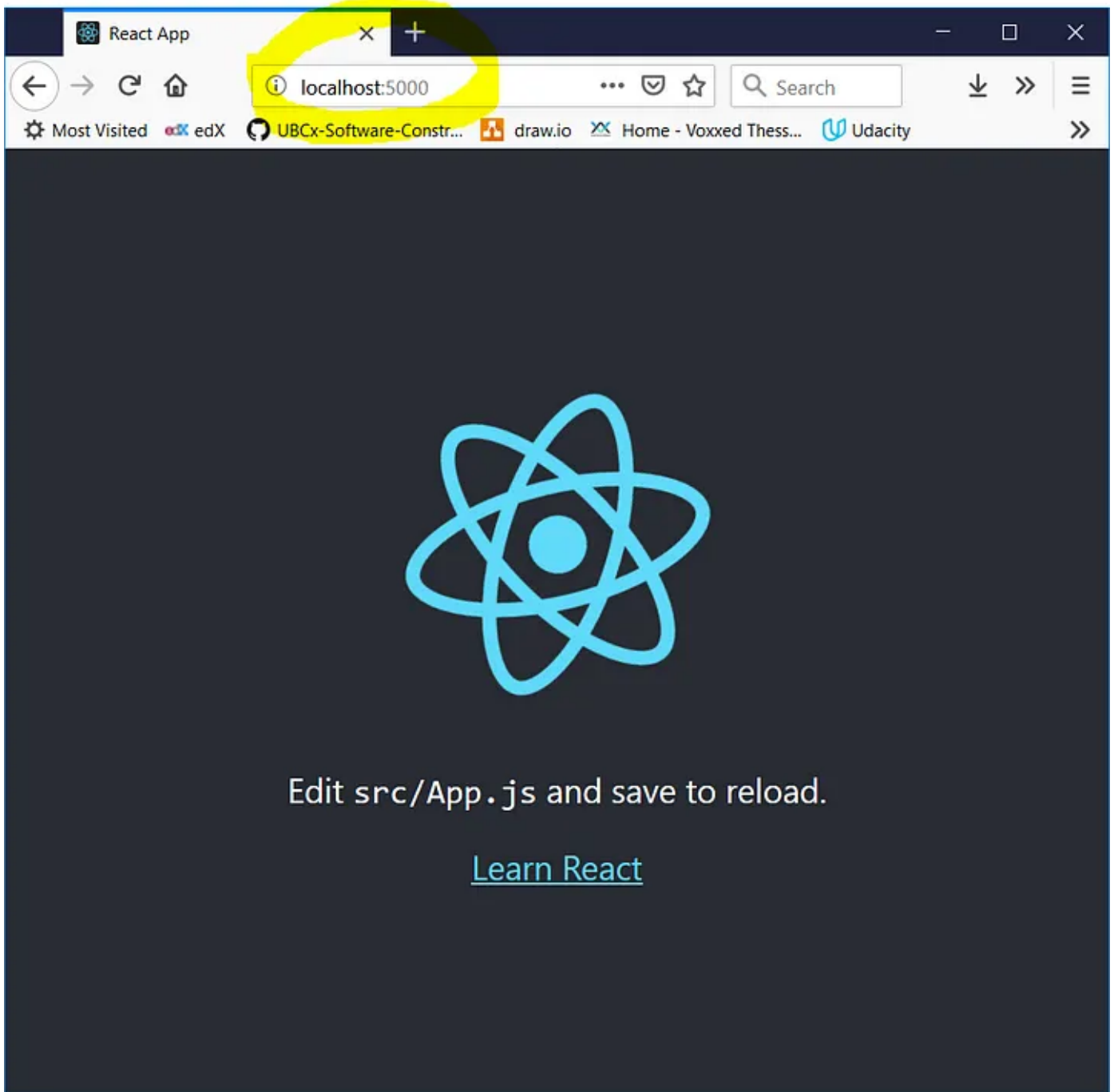
1: node

PS C:\Users\Achilles\Documents\Projects\Blog\server> node index.js
server started on port: 5000

A simple production server with node and express

in your browser go to

<http://localhost:5000>



Production server is working as expected on port 5000

Congratulations! Now your react app is served from the production server.

Optional : Deployment to heroku

if you want to deploy to Heroku add the following on the package.json

```
"engines": {  
  "node": "8.1.1",
```

```
"npm": "5.0.3"

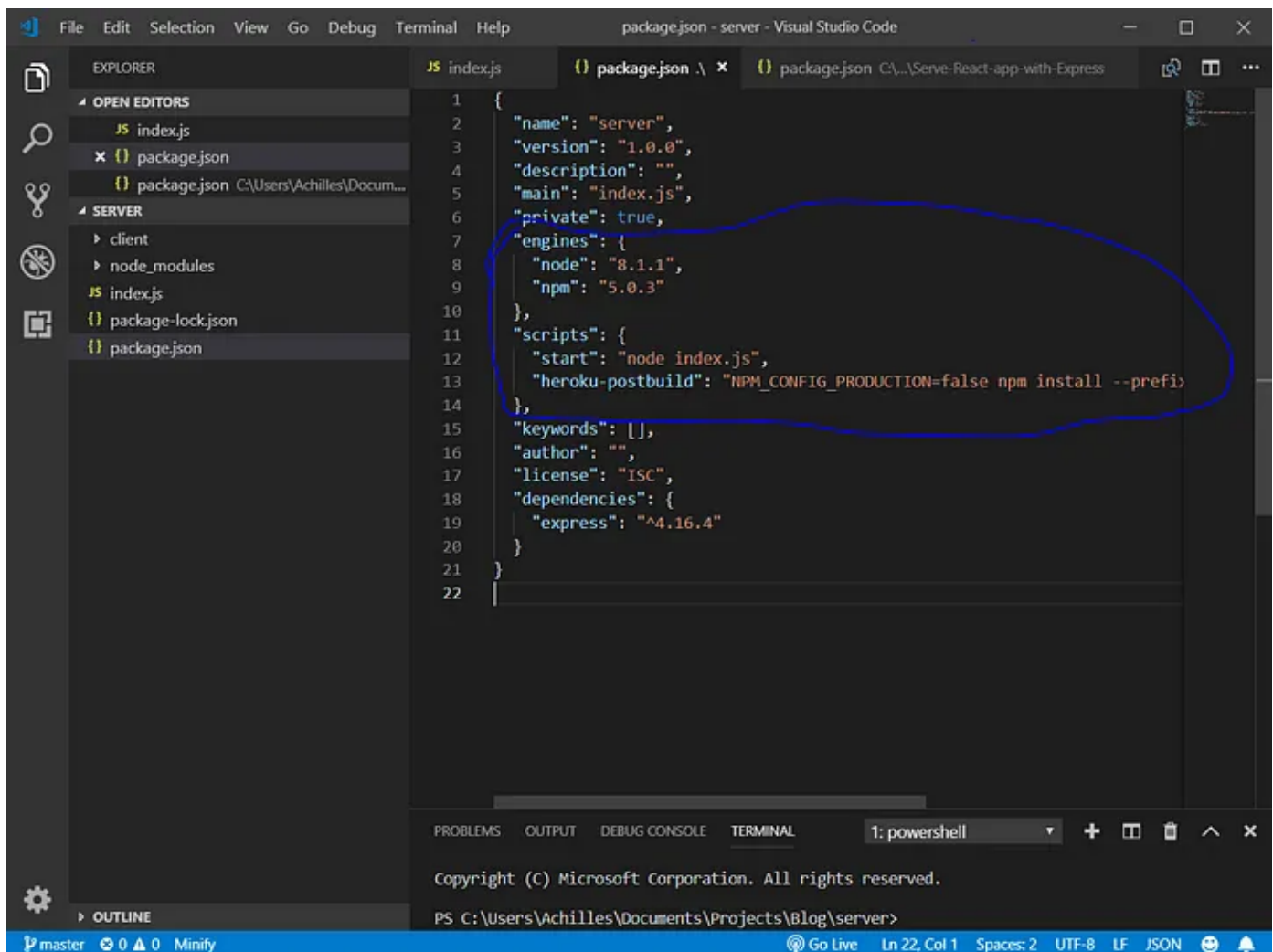
},

"scripts": {

"start": "node index.js",

"heroku-postbuild": "NPM_CONFIG_PRODUCTION=false npm install --prefix
client && npm run build --prefix client"

}
```



package.json for Heroku deployment

Now you can deploy your project on Heroku!

The following commands are executed by Heroku **after** it installs your **server dependencies** from **server/package.json** to install your react app dependencies.

This way you avoid committing to Heroku the build folder in the client directory and simply let Heroku build the project :)

```
NPM_CONFIG_PRODUCTION=false npm install --prefix client && npm run build --prefix client
```

- *-prefix client instructs npm to use the package.json in the client directory instead of the package.json in the server directory.*

NPM_CONFIG_PRODUCTION=false sets the Heroku npm to development mode (only for this command) to allow the installation of the dev dependencies necessary for the react app (client)

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Happy coding!

JavaScript

Expressjs

Nodejs

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Tech

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